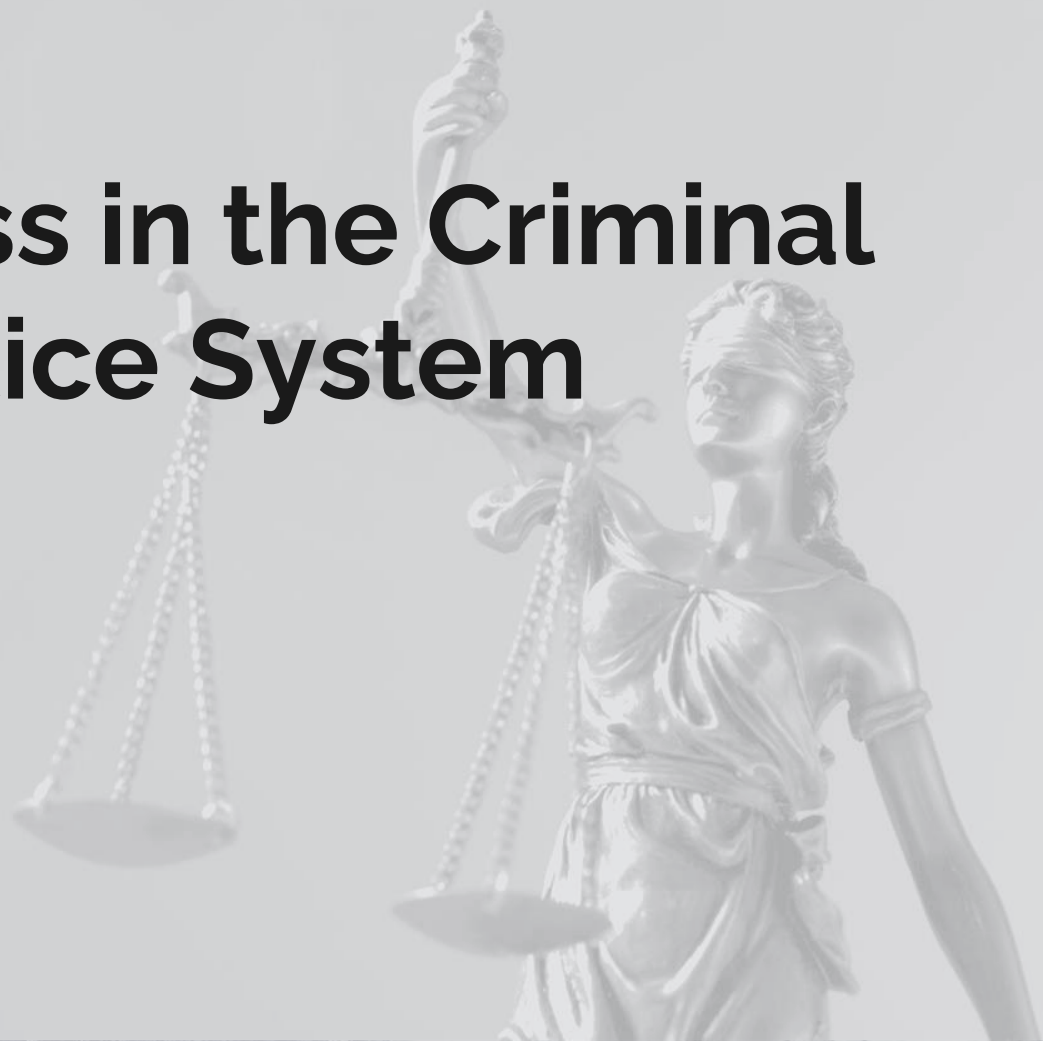


AI Fairness in the Criminal Justice System

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Junke Zhao



As AI takes on high-stakes decisions, analyzing fairness and bias is essential to ensure just outcomes

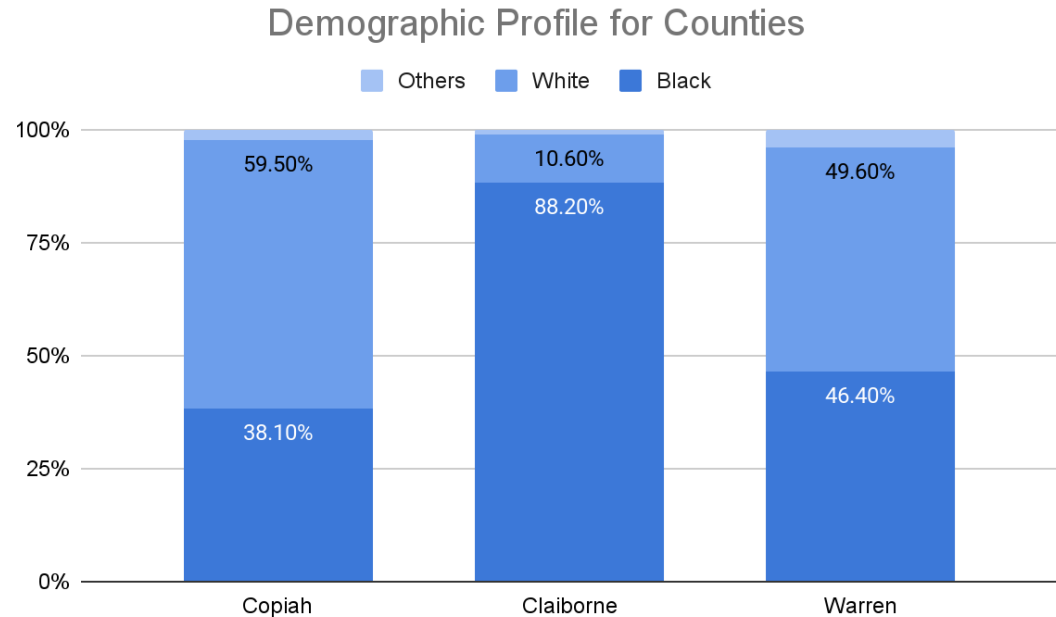


- AI is being increasingly used to offload mental work at increasingly higher stakes, such as with granting or denying bail
- This research is a culmination of our work and analyses conducted on 3 different county's synthetic data
- As we become increasingly reliant on AI, there will be a greater need to conduct analyses such as these to assess fairness and bias within these systems and address the problems they present

The three counties have different proportions of black, white and other races



- Racial distribution across the counties are largely different
- Copiah is mostly white
- Claiborne is mostly black
- Warren has a fairly even mix between the two predominant groups



Apart from race, the distribution across other features are similar across counties



- Outside of race, all counties had very similar distributions (within 1-3%) in:
 - Gender
 - Highest level of education completed
 - Economic status
 - Unemployment status
 - Income level
- In theory, these counties *should* have equal impact when using the AI-based risk scores across all individuals when determining to grant or deny bail



However...

Risk scores vary across different races and are the highest for the black population



- Assigned risk scores across racial demographics were very dissimilar
- Average risk score per race:
 - White: 4.73
 - Other: 4.73
 - Black: **5.86** - over a *full point* higher!
- Despite being similar across all other relevant factors, the black population received a higher risk score on average, which may imply the algorithm has trained on historical data

Judges' decision are more biased against the black population, showing disparate treatment

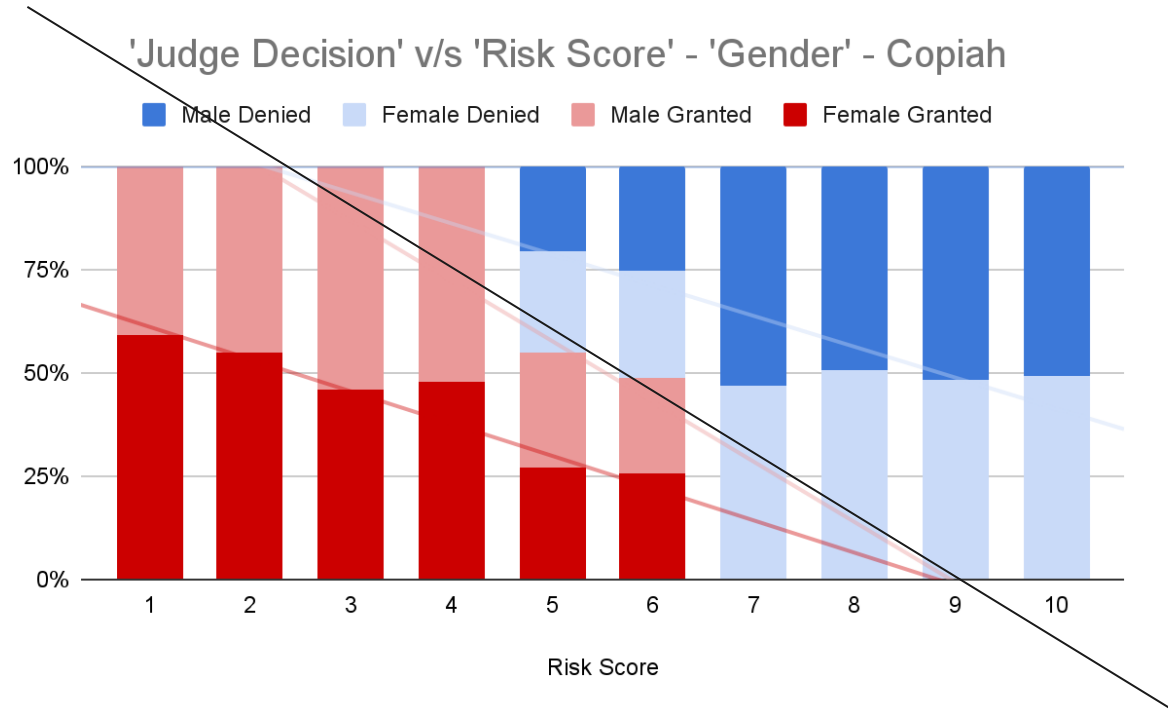
- Judge decision by race trends similarly across all counties.
- **Bail decisions** for black are **denied** beginning **risk score = 4** for **black** populations v/s
- Bail decisions for white are denied beginning risk score = 6 for white populations
- This displays a disparate treatment

Risk Score' Average by 'Gender'			
	Female	Male	Grand Total
Average of Risk Score	5.782	5.729	5.756

The judicial decisions across genders are symmetrical

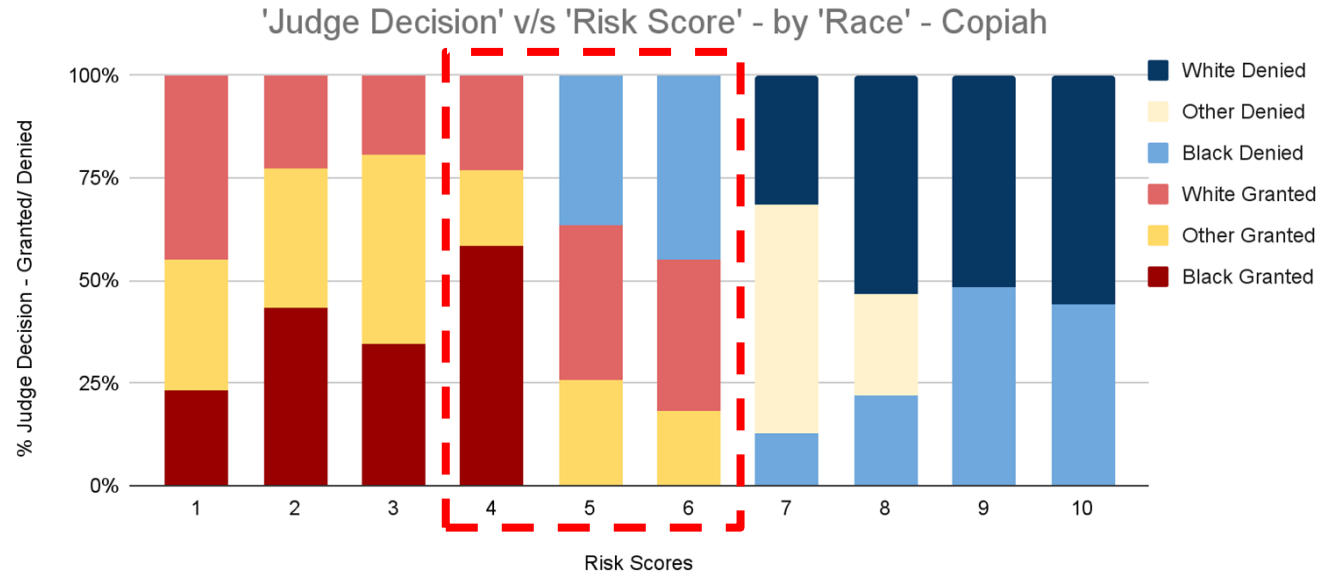


- This **black marked line** shows **symmetric treatment of two different demographic groups**.
- This can be considered as **normal, equal treatment by judges**.
- Copiah is an example, but this trend is common across all counties.



Judicial decisions across races have disparity

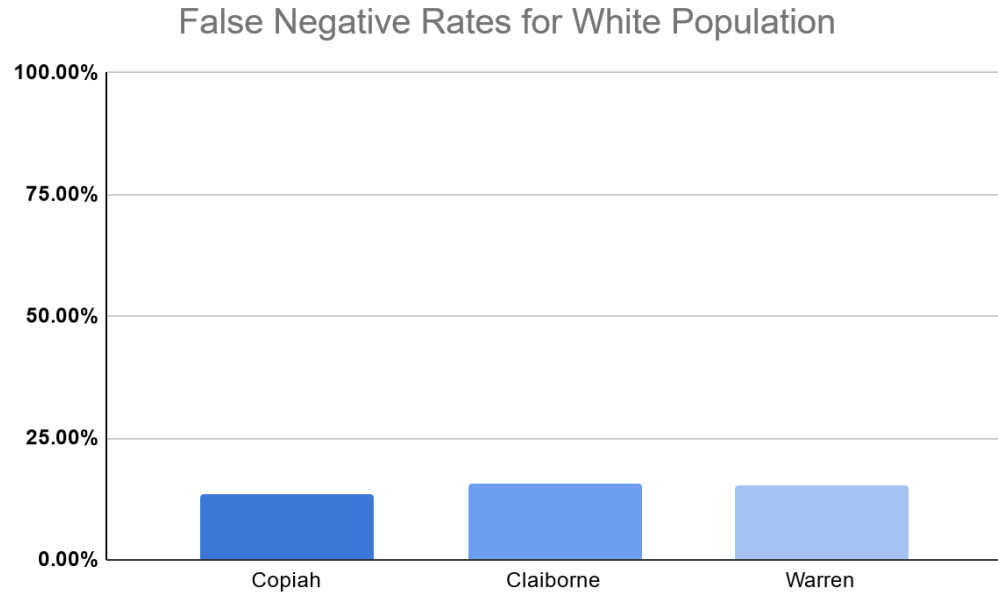
- No symmetry in diagram shows a **disparate treatment against black populations.**
- **Deviates from a normal**, equally representative treatment.
- Copiah is an example, but this trend is common across all counties.



Low FNR among the 'white' population



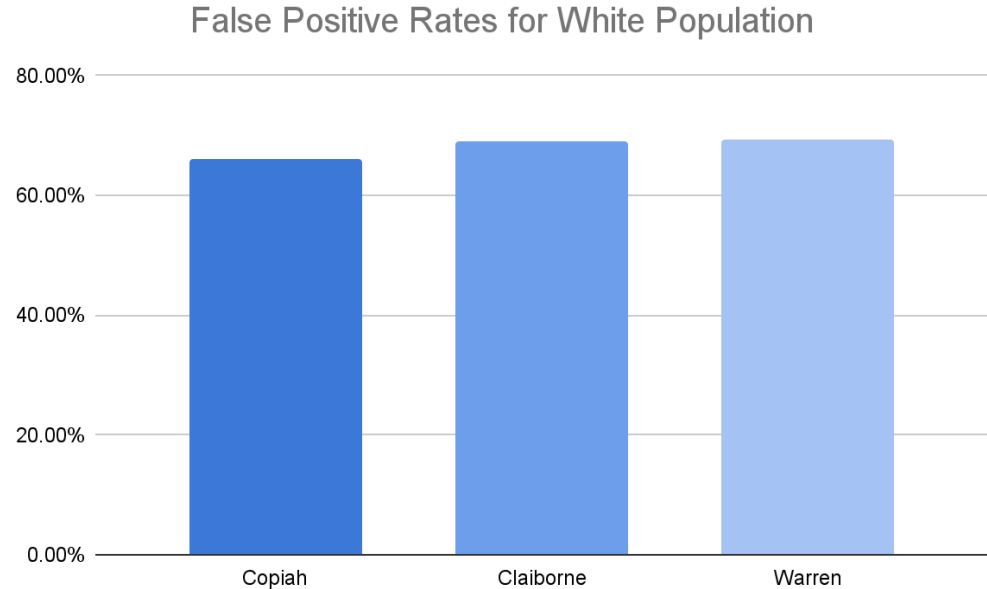
- The false negative rate for the white population is very low
- Few white individuals are denied bail that will not re-offend
- Implies the model is very accurate for this population



High FPR Highlights Model's Weakness in Predicting Re-Offenders for White Population



- Unlike the false negative rate, the false positive rate is very high
- White individuals are often mistakenly given bail when they will re-offend
- The model is inherently poor at identifying re-offenders

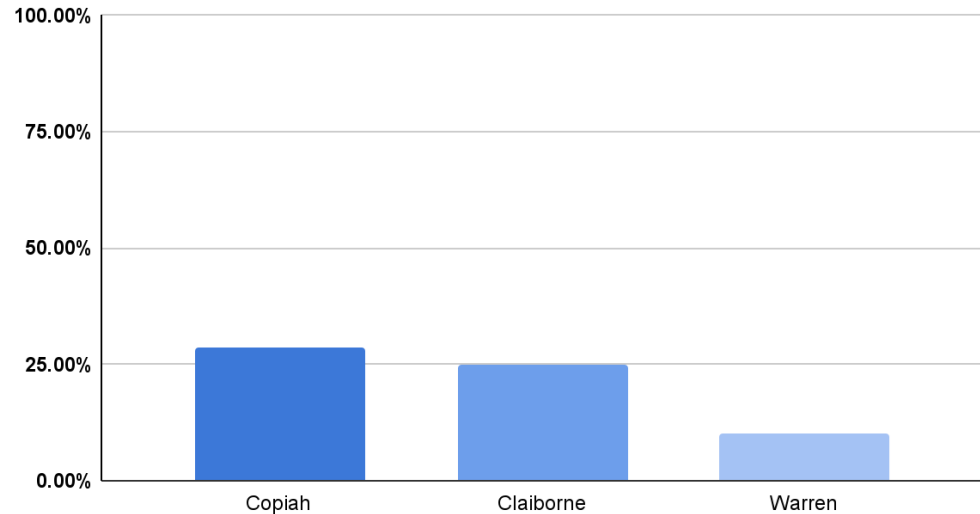


FNR for 'Other' Population: Comparable to White Demographic but Vary by County



- The 'other' population is very small by comparison
- The false negative rate is not very high
- Copiah and Claiborne are higher than Warren due to their smaller population
- Overall, relatively comparable to the white population

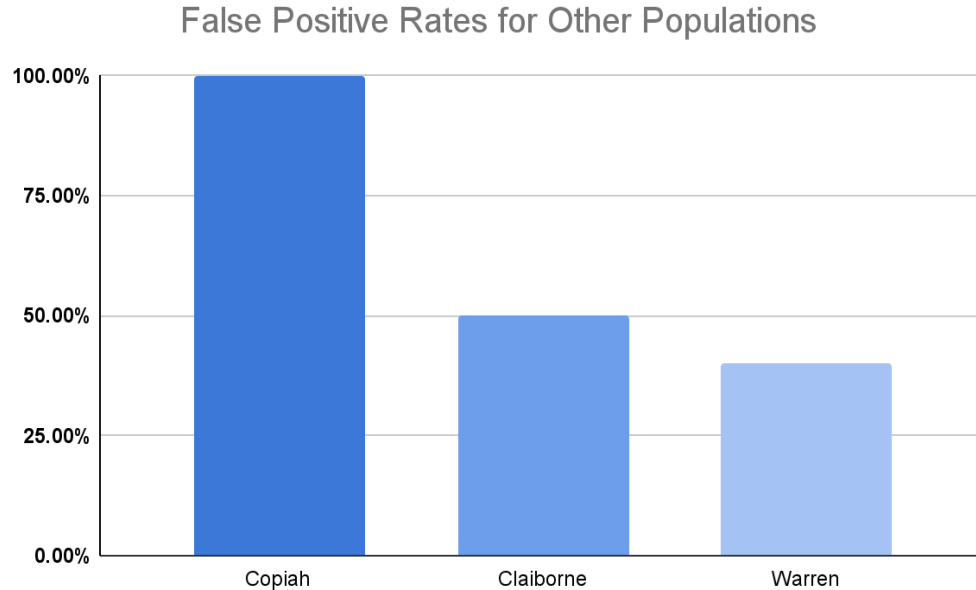
False Negative Rates for Others



Skewed FPR for 'Other' Populations Due to Small Sample Size



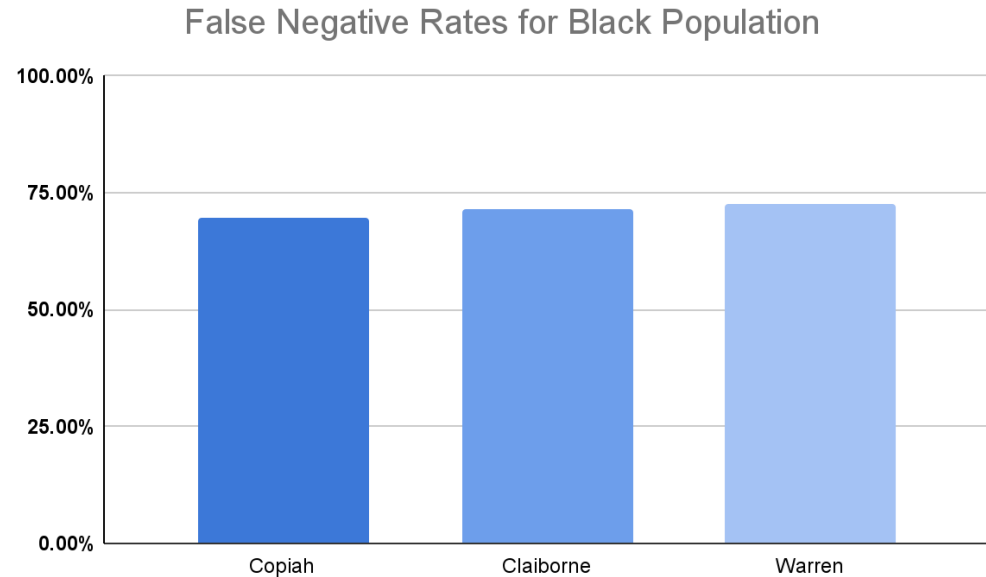
- The data for the 'other' population is skewed due to small population size
- Copiah has the smallest portion of this demographic



High FNRs: Model Bias for Black Population

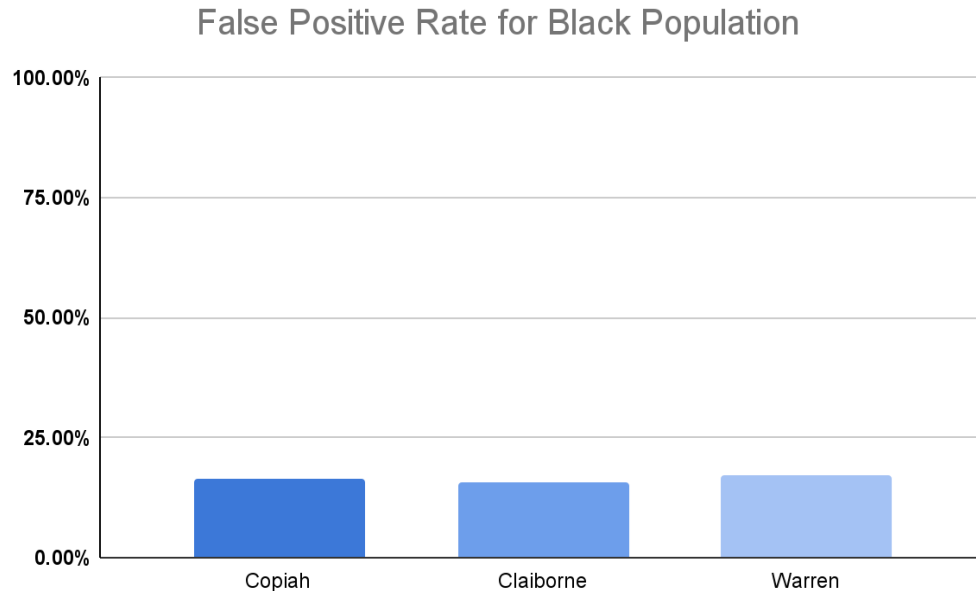


- Unlike the white and other populations, the FNR for the black population was much, much higher
- Up over 50 percent for all counties when compared to the white population
- This shows that the model is very poor at accurately identifying individuals who will re-offend



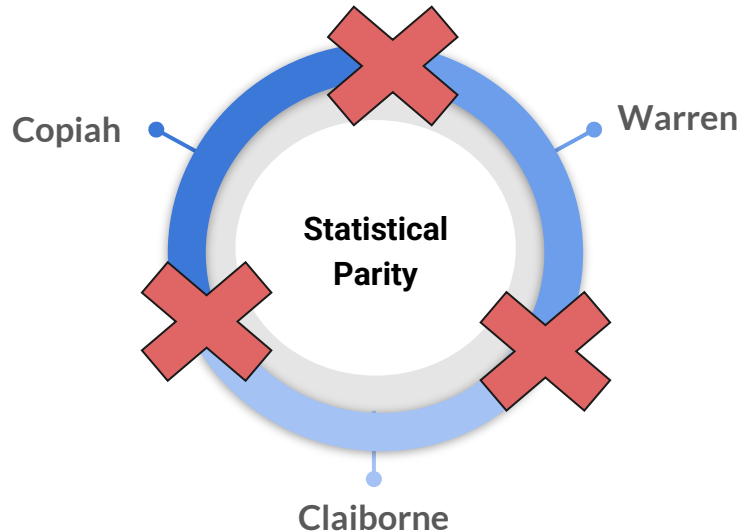
Low FPR Reveal Model's Failure to Identify Re-Offenders in Black Population

- The false positive rate for the black population is very low
- The model does a very poor job at identifying individuals who will re-offend
- This results in many black individuals being jailed unfairly

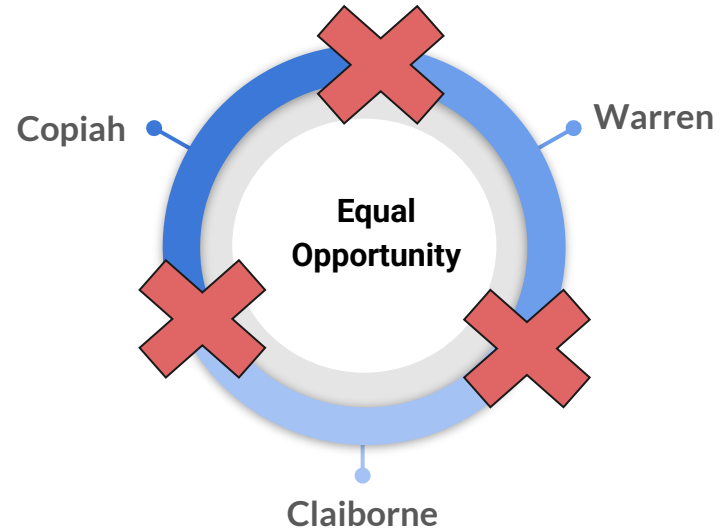


The model fails on both fairness metrics: statistical parity and equal opportunity

Is the rate of bails granted *independent* of which county an individual belongs to?



Is the rate of bails granted to non-reoffenders *consistent* across counties?



Biased FNR and FPR rates reveal unfair outcomes, with both model and judicial decisions lacking equity across groups



- **FNR is high for black populations** irrespective of the county, which means that for a black person, they are more likely to be jailed when they don't re-offend
- **FPR is high for white populations** irrespective of the county, which means that for a white person, they are more likely to be given bail when they will re-offend
- The model is not the only thing that is biased – **judge decisions are also biased**, as reflected in the data
- **No statistical parity or equal opportunity** across any of the counties, which implies that the outcomes are not fair across population segments

Thank you!



Collaboration Log



Assembly and folder setup - Katelyn

Data analysis framework - Katelyn

Data analysis edits - Vashishth, Ashwin

Copiah County - Ashwin, Vashishth

Claiborne County - Katelyn

Warren County - Satish, Junke

Essay - Vashishth, Ashwin, Katelyn

Essay editing - Katelyn

Presentation - Vashishth, Ashwin, Satish, Katelyn